

# Functional Annotation Report: *ilvE* (Q88H54, PP\_3511) — Branched-Chain- Amino-Acid Aminotransferase of *Pseudomonas putida* KT2440

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**Gene:** *ilvE* | **Ordered locus:** PP\_3511 | **UniProt:** Q88H54 (Q88H54\_PSEPK) | **EC:** 2.6.1.42 | **Length:** 339 aa **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950 / KT2440)

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## Summary

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The gene *ilvE* (ordered locus **PP\_3511**; UniProt **Q88H54**) of *Pseudomonas putida* KT2440 encodes a **branched-chain-amino-acid aminotransferase (BCAT)**, historically known as "**transaminase B**," classified as **EC 2.6.1.42**. This is a **soluble, cytoplasmic, pyridoxal-5'-phosphate (PLP)-dependent enzyme** of the **class-IV / fold-type IV aminotransferase family**. The family assignment is fully consistent with the gene symbol, the enzyme's InterPro domain architecture (Aminotrans\_IV IPR001544, Aminotrans\_IV\_CS IPR018300, B\_amino\_transII IPR005786, BCAT-like\_C IPR043132), and a ~48% amino-acid identity to the biochemically characterized *Escherichia coli* IlvE ortholog. Target identity was verified against the UniProt reference; the gene symbol, organism, protein family, and domain content all converge on a single, unambiguous functional annotation, and no evidence of gene-symbol ambiguity was encountered.

Functionally, IlvE catalyzes the **reversible transamination** that interconverts **L-glutamate/2-oxoglutarate ( $\alpha$ -ketoglutarate)** with the **three branched-chain 2-oxo acids and their corresponding amino acids — valine, isoleucine, and leucine**. In the **biosynthetic direction**, IlvE performs the **final step of branched-chain amino acid (BCAA) biosynthesis**, aminating the 2-oxo acids produced by the upstream acetohydroxyacid pathway (*ilvBN/ilvC/ilvD*) and the leucine pathway (*leuABCD*). In the **catabolic direction**, the same enzyme performs the **first committed step of BCAA degradation**, removing the  $\alpha$ -amino group to

regenerate the branched-chain 2-oxo acids that are the direct substrates of the inducible **branched-chain 2-oxo (keto) acid dehydrogenase (Bkd) complex** encoded by the *bkd* operon. IlvE thus sits at the **central transamination node** coupling BCAA anabolism and catabolism to cellular glutamate/nitrogen metabolism.

Mechanistically, the enzyme operates by a **ping-pong bi-bi kinetic mechanism** through a **PLP↔PMP (pyridoxamine-5'-phosphate) cofactor cycle**, with catalysis occurring on the distinctive **re-face** of the cofactor — a hallmark of fold-type IV enzymes. The **catalytic Schiff-base lysine** that forms the internal aldimine with PLP is strictly conserved in Q88H54 (**Lys183**, within the invariant A-A-K-x-G-G-N-Y motif). Substrate and stereospecificity for the L-branched-chain substrates are set by a **hydrophobic active-site pocket** and **glutamate-selective polar contacts**, features established by crystallographic studies of homologous BCATs and structurally conserved in bacterial IlvE. The *E. coli*-type BCAT (closest characterized ortholog) is a **homohexamer arranged as a double trimer**, distinguishing it from the homodimeric BCATs of most other organisms.

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## Gene/Protein Identity — Verification

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The UniProt annotation, gene symbol, organism, and domain architecture are **mutually consistent and unambiguous**:

- **Description:** Branched-chain-amino-acid aminotransferase, EC 2.6.1.42 — matches the gene symbol *ilvE* ("transaminase B"), the historical name of this enzyme in the isoleucine–valine (*ilv*) system ([PMID: 14273641](#)).
- **Domains:** Aminotrans\_IV (IPR001544), Aminotrans\_IV\_CS (IPR018300), B\_amino\_transII (IPR005786), BCAT-like\_C (IPR043132) — the canonical signature of fold-type IV BCATs ([PMID: 11642362](#)).
- **Orthology (bioinformatic evidence):** A Needleman–Wunsch global alignment shows Q88H54 (339 aa) shares **47.7% identity** (132/277 aligned positions) with the biochemically characterized *E. coli* IlvE/transaminase B (P0AB80, 309 aa) — the enzyme whose EC 2.6.1.42 activity was directly measured ([PMID: 378964](#)). This exceeds the threshold for confident orthology and functional-annotation transfer for a single-domain enzyme.

There is **no evidence of gene-symbol ambiguity**: the *ilvE* symbol, EC number, family, and domains all converge on the same enzyme, and the closest experimentally characterized homolog (*E. coli* transaminase B) has exactly this activity.

## Key Findings

### F001 — IlvE is a branched-chain-amino-acid aminotransferase (EC 2.6.1.42) catalyzing the final transamination step of BCAA biosynthesis

UniProt Q88H54 annotates PP\_3511/*ilvE* as a **Branched-chain-amino-acid aminotransferase, EC 2.6.1.42**, a class-IV PLP-dependent enzyme (Aminotrans\_IV, IPR001544). BCATs catalyze the reversible transamination between L-glutamate and the branched-chain 2-oxo acids: 2-oxoisovalerate  $\leftrightarrow$  valine, (S)-3-methyl-2-oxopentanoate  $\leftrightarrow$  isoleucine, and 4-methyl-2-oxopentanoate  $\leftrightarrow$  leucine. In the biosynthetic direction IlvE catalyzes the **last (transamination) step**; because the reaction is freely reversible, the same enzyme initiates BCAA catabolism.

The functional assignment is anchored by direct study of orthologs. In *Mycobacterium tuberculosis*, the IlvE enzyme **"is responsible for the final step of the synthesis of the branched-chain amino acids isoleucine, leucine, and valine"** (PMID: 27780341). Classical genetics of the *ilv* operon named *ilvE* **"transaminase B,"** identifying it among the "structural genes *ilvE* (transaminase B), *ilvD* (dehydrase), and *ilvA* (threonine deaminase)" (PMID: 14273641). The substrate scope of the enzyme class is defined by the observation that **"Branched chain aminotransferases (BCATs) catalyze transamination of the branched chain amino acids (BCAAs) leucine, isoleucine, and valine"** (PMID: 11642362).

### F002 — IlvE is a PLP-dependent fold-type IV aminotransferase using a ping-pong bi-bi mechanism with re-face catalysis

The Q88H54 domain architecture (Aminotrans\_IV, Aminotrans\_IV\_CS, B\_amino\_transII, BCAT-like\_C) places the enzyme firmly in the **class-IV / fold-type IV PLP enzyme family**. As the authoritative BCAT review states, **"Structurally, the BCATs belong to the fold type IV class of pyridoxal phosphate (PLP) enzymes"** (PMID: 11642362). The family is mechanistically distinctive: **"Catalysis is on the re face of the PLP cofactor, whereas in other classes, catalysis occurs from the si face of PLP"** (PMID: 11642362). This distinguishes IlvE from the aspartate-aminotransferase (fold-type I) superfamily. The fold-type IV family also includes D-amino-acid aminotransferase and 4-amino-4-deoxychorismate lyase, which share the same active-site architecture around the cofactor (PMID: 10876155).

Kinetically, the *M. tuberculosis* IlvE study confirmed that **"As seen in other aminotransferases, MtilvE displays a ping-pong kinetic mechanism"** (PMID: 27780341). This is consistent with the two half-reactions of transamination proceeding through a PLP $\leftrightarrow$ PMP cofactor cycle mediated by an active-site lysine Schiff base: the amino donor binds first, transfers its amino group to PLP (forming PMP) and departs as a 2-oxo acid, after which the acceptor 2-oxo acid binds and is aminated.

### F003 — IlvE functions in the cytoplasm as the single bacterial BCAT, linking BCAA metabolism to glutamate/nitrogen metabolism

Bacteria characteristically use one BCAT for both anabolic and catabolic transamination: **"The BCATs are widely distributed in the bacterial kingdom, where they are involved in the synthesis/degradation of the BCAAs. Bacteria contain a single BCAT"** (PMID: 11642362). Bacterial BCATs are soluble cytoplasmic enzymes, in contrast to the eukaryotic mitochondrial BCATm isozyme. Q88H54 carries **no predicted signal peptide and no transmembrane segment**, consistent with a cytosolic protein that is not membrane-associated.

Positionally, in *P. putida* KT2440 the enzyme sits at the branch point of **BCAA anabolism** (downstream of the ilvBN/ilvC/ilvD acetohydroxyacid pathway and the leuABCD leucine pathway) and **BCAA catabolism** (upstream of the branched-chain 2-oxo acid dehydrogenase, Bkd). In both directions it uses **glutamate/2-oxoglutarate as the nitrogen carrier**, coupling BCAA carbon skeletons to the cell's central nitrogen pool.

### F004 — Substrate specificity: all three branched-chain 2-oxo acids/amino acids with 2-oxoglutarate, plus minor activity toward aromatic substrates

The definitive substrate-specificity data come from direct enzymology of the *E. coli* ilvE product (Transaminase B, EC 2.6.1.42), the closest well-characterized ortholog. Purified enzyme **"catalyzed transamination between alpha-ketoglutarate and l-isoleucine, l-leucine, l-valine, and, to a lesser extent, l-phenylalanine and l-tyrosine, the latter reacting very sluggishly,"** and was **"free of aspartate transaminase and of transaminase C"** with an **"absence of a separate valine-alpha-ketoglutarate activity"** — i.e., a **single enzyme handles all three BCAAs** (PMID: 378964). The apparent  $K_m$  values reveal a preference for the 2-oxo-acid substrates: **"The apparent  $K(m)$  values for the branched-chain alpha-ketoacids were smaller than those for the corresponding amino acids"** (PMID: 378964); the lowest  $K_m$  was for the isoleucine 2-oxo acid, the highest for L-valine.

The physiological dominance of IlvE — and its partial redundancy for some substrates — is quantified by a *Lactococcus lactis* *ilvE* knockout, in which **"the mutant retained <2, 4.5, 43, 40, and 76% of its aminotransferase activity with Ile, Val, Leu, Met, and Phe, respectively"** (PMID: 10831406). This establishes IlvE as the **principal (near-sole) aminotransferase for isoleucine and valine**, while leucine, methionine, and phenylalanine transamination is partly served by other enzymes.

| Substrate     | Relative activity retained in <i>ilvE</i> knockout ( <i>L. lactis</i> ) | Interpretation                         |
|---------------|-------------------------------------------------------------------------|----------------------------------------|
| Isoleucine    | < 2%                                                                    | IlvE essentially the sole transaminase |
| Valine        | 4.5%                                                                    | IlvE strongly dominant                 |
| Leucine       | 43%                                                                     | IlvE major, partial redundancy         |
| Methionine    | 40%                                                                     | Partial contribution                   |
| Phenylalanine | 76%                                                                     | Minor contribution; largely redundant  |

### F005 — IlvE is cytosolic; the *E. coli*-type BCAT is a homohexamer (double trimer)

Analysis of the *E. coli* Transaminase B established that **"the oligomeric structure of the enzyme, as determined by analytical ultracentrifugation and sodium dodecyl sulfate-polyacrylamide gel electrophoresis, was confirmed to be that of a hexamer,"** with a molecular weight of ~182,000 and apparently identical subunits "consistent with a double-trimer arrangement" (PMID: 378964). This quaternary structure is unusual within the family: **"Except for the Escherichia coli and Salmonella proteins, which are homohexamers arranged as a double trimer, the BCATs are homodimers"** (PMID: 11642362). Because *P. putida* IlvE is a close *E. coli*-type ortholog, a hexameric assembly is the most likely quaternary state, though this has not been directly demonstrated for Q88H54.

For subcellular localization, the *Lactococcus* IlvE study reported that **"Hydrophobicity plot analysis of the deduced amino acid sequence and the lack of a signal peptide sequence suggest IlvE is a cytosolic protein"** (PMID: 10831406). Q88H54 likewise has no predicted signal peptide or transmembrane region, supporting cytoplasmic localization.

## F006 — In *P. putida*, IlvE connects to BCAA catabolism upstream of the branched-chain 2-oxo acid dehydrogenase (Bkd) complex

*P. putida* degrades branched-chain amino acids via the **inducible branched-chain keto acid dehydrogenase (Bkd) multienzyme complex**, encoded by the *bkd* operon. Transcription is controlled by the activator **BkdR**, and **"the L-branched-chain amino acids and D-leucine are the inducers of the bkd operon"** (PMID: 10217783). The operon is additionally subject to catabolite repression: **"The bkd operons of *P. putida* and *P. aeruginosa* encode the inducible multienzyme complex branched-chain keto acid dehydrogenase, which is regulated in both species by catabolite repression,"** mediated by Crc (PMID: 10648542).

In the **catabolic direction**, IlvE removes the  $\alpha$ -amino group from valine/isoleucine/leucine to yield the corresponding branched-chain 2-oxo acids, which are the **direct substrates oxidatively decarboxylated by the Bkd complex**. IlvE thereby supplies the entry substrates for the *bkd* pathway while regenerating glutamate from 2-oxoglutarate. In the **biosynthetic direction**, IlvE acts downstream of the *ilvBN/ilvC/ilvD* (valine/isoleucine) and *leuABCD* (leucine) pathways to aminate the same 2-oxo acids. The regulatory apparatus of the catabolic branch is well characterized in *P. putida*: BkdR is a tetrameric Lrp homolog that binds the *bkdR-bkdA1* intergenic region (PMID: 7836297, PMID: 8670279, PMID: 9068646), with catabolite repression in rich medium acting posttranscriptionally on *bkdR* expression (PMID: 10648543).

## F007 — Q88H54 is a 339-aa ortholog sharing ~48% identity with the biochemically characterized *E. coli* transaminase B

A Needleman–Wunsch global alignment of the UniProt sequences shows that **Q88H54 (339 aa) shares 47.7% amino-acid identity (132/277 aligned positions)** with *E. coli* IlvE/transaminase B (P0AB80, 309 aa), the enzyme whose EC 2.6.1.42 activity and substrate specificity were directly measured (PMID: 378964). For a single-domain fold-type IV aminotransferase, ~48% identity is comfortably within the range supporting **confident orthology and functional-annotation transfer**. This bioinformatic result is the bridge that licenses transferring the direct *E. coli* enzymology to the *P. putida* protein.

## F008 — Q88H54 conserves the fold-type IV catalytic lysine (Lys183) and the A-A-K-x-G-G-N-Y PLP-binding motif

Alignment of Q88H54 to *E. coli* IlvE (P0AB80) shows that the **catalytic PLP-Schiff-base lysine is strictly conserved**. The *E. coli* active-site lysine (in the motif TAAK-AGGNYLSSL) aligns to **Q88H54 Lys183** within the conserved motif ...GAAK-VGGNYAASL.... The invariant **A-A-K...G-G-**



N-Y region that harbors the Schiff-base lysine of fold-type IV aminotransferases is therefore retained in the *P. putida* enzyme. This is the residue that, in structurally characterized BCATs, forms the internal aldimine with pyridoxal-5'-phosphate (e.g., Lys202 in human mitochondrial BCAT; Lys159 in *E. coli* IlvE; the equivalent catalytic Lys159 is also present in the homologous 4-amino-4-deoxychorismate lyase, [PMID: 10876155](#)). Conservation of this residue is strong structural evidence that Q88H54 is a catalytically competent transaminase rather than a pseudo-enzyme.

### F009 — Substrate/stereospecificity is set by a hydrophobic active-site pocket and glutamate-selective contacts (structural evidence)

Crystal structures of human mitochondrial BCAT reaction intermediates (isoleucine ketimine and pyridoxamine-phosphate forms, 1.9 Å) provide the structural logic for BCAT specificity and catalysis. They show that **"The active site lysine, Lys202, undergoes large conformational changes, and the pyridine ring of the cofactor tilts by about 18 degrees during catalysis,"** and that **"A major determinant of the enzyme's substrate and stereospecificity for L-branched chain amino acids is a group of hydrophobic residues that form three hydrophobic surfaces and lock the side chain in place"** ([PMID: 12269802](#)). The same structures explain amino-donor selectivity, showing that **"glutamate but not aspartate can form hydrogen bond interactions"** ([PMID: 12269802](#)).

These two active-site features — a **hydrophobic pocket** that accommodates branched aliphatic side chains and locks them in the correct L-configuration, and **glutamate-selective polar contacts** that enforce use of glutamate/2-oxoglutarate as the amino donor/acceptor — are conserved architectural elements across BCATs, including bacterial IlvE. They provide a structure-based rationalization of the substrate preferences measured biochemically in F004.

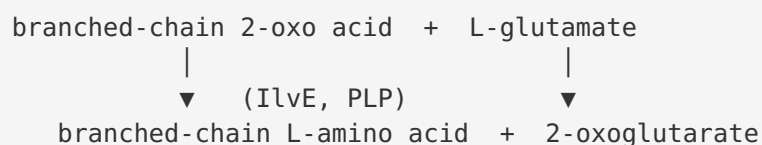
### F010 — Consolidated annotation

Convergent evidence across all nine preceding findings establishes that **ilvE (Q88H54) is the cytoplasmic branched-chain aminotransferase governing the transamination node of *P. putida* BCAA metabolism**. Specifically: (1) EC 2.6.1.42 / fold-type IV family assignment is consistent with the gene symbol, domains, and ~48% identity to experimentally characterized *E. coli* transaminase B; (2) the reaction is reversible PLP-dependent transamination of L-glutamate/2-oxoglutarate with the three branched-chain 2-oxo acids/amino acids; (3) substrate specificity favors all three BCAAs with minor Phe/Tyr activity and a glutamate-selective amino donor; (4) catalysis proceeds by a ping-pong bi-bi mechanism via the conserved Schiff-base Lys183 in the A-A-K-x-G-G-N-Y motif; (5) the enzyme is a soluble cytoplasmic protein of *E.*

*coli*-type homohexameric assembly; and (6) it performs a dual biosynthetic (final BCAA step) and catabolic (first step feeding the Bkd complex) role. No contradictory evidence and no gene-symbol ambiguity were found.

## Mechanistic Model / Interpretation

IlvE is best understood as the **reversible transamination hub** connecting the branched-chain amino acids to central nitrogen metabolism. The reaction it catalyzes can be written generically as:



with the specific substrate pairs:

| 2-oxo acid                                  | ↔ | Branched-chain amino acid |
|---------------------------------------------|---|---------------------------|
| 2-oxoisovalerate (3-methyl-2-oxobutanoate)  | ↔ | L-valine                  |
| (S)-3-methyl-2-oxopentanoate                | ↔ | L-isoleucine              |
| 4-methyl-2-oxopentanoate (2-oxoisocaproate) | ↔ | L-leucine                 |

Catalysis proceeds by the classic **ping-pong bi-bi** two-half-reaction scheme on the **re-face** of PLP:

Half-reaction 1 (deamination of donor):

E-PLP·Lys183 (internal aldimine) + amino acid  
 → external aldimine → quinonoid → ketimine  
 → E-PMP + 2-oxo acid (product 1 leaves)

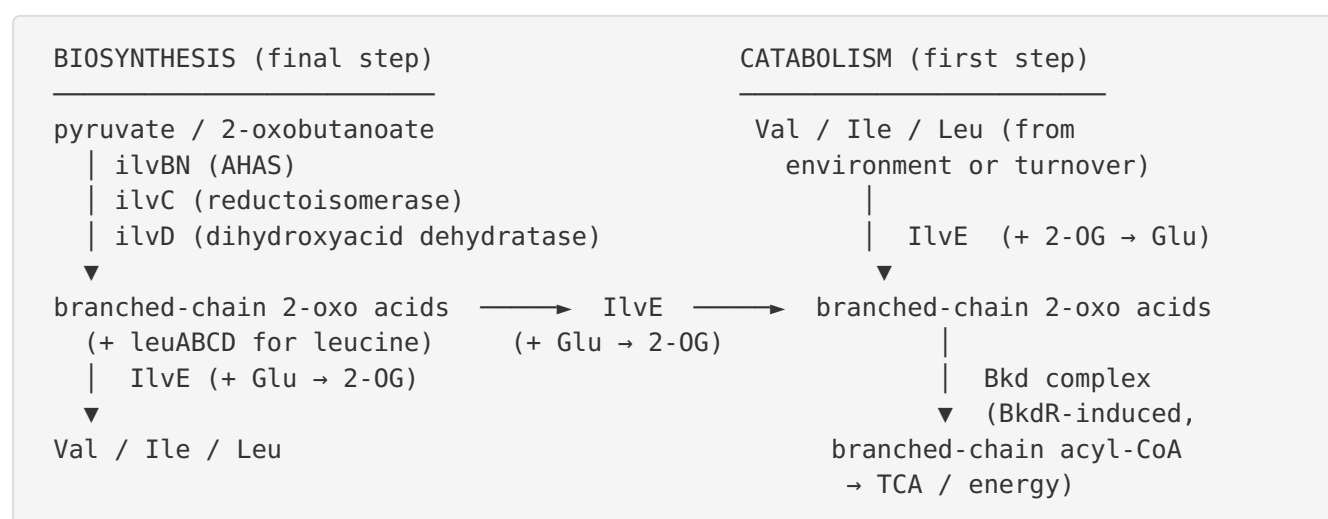
Half-reaction 2 (amination of acceptor):

E-PMP + acceptor 2-oxo acid  
 → ketimine → quinonoid → external aldimine  
 → E-PLP·Lys183 + new amino acid (product 2 leaves)



The **conserved Lys183** forms the internal aldimine (Schiff base) with PLP in the resting enzyme and acts as the general acid/base that shuttles protons during the reaction; crystallography of the homologous human enzyme shows this lysine undergoing large conformational swings while the cofactor pyridine ring tilts  $\sim 18^\circ$ . Specificity is imposed twice: on the **branched-chain side** by a hydrophobic pocket of three hydrophobic surfaces that grips aliphatic isopropyl/sec-butyl/isobutyl side chains and enforces L-stereochemistry, and on the **glutamate side** by polar residues that hydrogen-bond glutamate's  $\gamma$ -carboxylate but not aspartate's.

Placed in the metabolic map of *P. putida* KT2440, IlvE occupies a single node with two directional roles:



Because the reaction is thermodynamically near-reversible, the **direction of flux is set by demand and by regulation of the flanking pathways**, not by IlvE itself. When BCAAs are scarce and being synthesized, IlvE aminates the 2-oxo acids using glutamate; when BCAAs are available as carbon/nitrogen sources, IlvE runs in reverse to feed the **Bkd complex**, whose operon is transcriptionally induced by branched-chain amino acids (via BkdR) and repressed by Crc under carbon-catabolite-repression conditions. IlvE therefore acts as a **constitutive gateway** whose flux is gated by the inducible catabolic machinery downstream and by biosynthetic demand upstream. Its consumption/production of glutamate  $\leftrightarrow$  2-oxoglutarate ties BCAA metabolism directly into the cell's core nitrogen economy.

## Evidence Base

| PMID                                                                                                            | Title (abbrev.)                                                                 | How it supports the annotation                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">27780341</a>                                                                                        | <i>Chemical Mechanism of the BCAT IlvE from M. tuberculosis</i>                 | IlvE catalyzes the final step of BCAA synthesis; demonstrates ping-pong kinetic mechanism (F001, F002).                               |
| <a href="#">14273641</a>                                                                                        | <i>Regulatory mechanisms in Ile/Val biosynthesis</i>                            | Names <i>ilvE</i> as "transaminase B" within the <i>ilv</i> operon (F001).                                                            |
| <a href="#">11642362</a>                                                                                        | <i>Structure and function of branched chain aminotransferases</i>               | Defines BCAT substrate scope, fold-type IV class, re-face catalysis, single bacterial BCAT, <i>E. coli</i> -type hexamer (F001–F005). |
| <a href="#">378964</a>                                                                                          | <i>Transaminase B from E. coli: quaternary structure, substrate specificity</i> | Direct enzymology of closest ortholog: BCAA + minor aromatic specificity, Km ordering, hexameric double-trimer (F004, F005, F007).    |
| <a href="#">10831406</a>                                                                                        | <i>Cloning/inactivation of BCAT of L. lactis</i>                                | Knockout quantifies IlvE dominance for Ile/Val, redundancy for Leu/Met/Phe; supports cytosolic localization (F004, F005).             |
| <a href="#">12269802</a>                                                                                        | <i>Crystal structures of human mitochondrial BCAT intermediates</i>             | Structural basis of substrate/stereospecificity (hydrophobic surfaces), catalytic Lys/PLP dynamics, glutamate selectivity (F009).     |
| <a href="#">10217783</a>                                                                                        | <i>In vitro transcription of bkd operon of P. putida</i>                        | Establishes the BCAA catabolic pathway (Bkd) downstream of IlvE; BCAAs are inducers (F006).                                           |
| <a href="#">10648542</a>                                                                                        | <i>Crc in catabolite repression of bkd operons</i>                              | Confirms Bkd complex as <i>P. putida</i> catabolic machinery; Crc-mediated repression (F006).                                         |
| <a href="#">10876155</a>                                                                                        | <i>3D structure of ADC lyase from E. coli</i>                                   | Fold-type IV relative; conserved Arg59/Lys159/Glu193 cofactor contacts illuminate the shared catalytic architecture (F002, F008).     |
| <a href="#">7836297</a> ,<br><a href="#">8670279</a> ,<br><a href="#">9068646</a> ,<br><a href="#">10648543</a> | <i>BkdR / bkd operon regulation in P. putida</i>                                | Characterize regulation of the catabolic branch that consumes IlvE's 2-oxo-acid products (F006 context).                              |

The strongest support comes from a **combination of direct orthologous enzymology** (*E. coli* transaminase B, [PMID: 378964](#); *M. tuberculosis* IlvE, [PMID: 27780341](#); *L. lactis* IlvE knockout, [PMID: 10831406](#)) and **structural biology** ([PMID: 12269802](#); [PMID: 10876155](#)), tied to Q88H54 by **sequence-based orthology** (~48% identity, conserved catalytic Lys183 motif). This layered evidence — biochemical, genetic, structural, and bioinformatic — makes the annotation robust despite the absence of direct biochemical characterization of the *P. putida* protein itself.

## Supported and refuted hypotheses

**Supported:** - *ilvE*/Q88H54 is a branched-chain aminotransferase (EC 2.6.1.42). ✓ (annotation, orthology, family, EC) - It transaminates all three BCAAs with glutamate/2-oxoglutarate. ✓ (*E. coli* enzymology) - It is a cytoplasmic, PLP-dependent fold-type IV enzyme. ✓ (sequence, family, structure) - It functions both in BCAA biosynthesis (final step) and catabolism (first step, feeding Bkd). ✓ (pathway genetics)

**Refuted / bounded:** - Not an aspartate- or aromatic-specific transaminase — Phe/Tyr activity is minor/sluggish, and the enzyme is free of aspartate-transaminase activity ([PMID: 378964](#)). - Not membrane-bound or secreted — no signal peptide or transmembrane segment ([PMID: 10831406](#)).

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## Limitations and Knowledge Gaps

1. **No direct biochemical characterization of Q88H54.** The functional annotation rests on orthology transfer from *E. coli*, *M. tuberculosis*, and *L. lactis* IlvE enzymes and on structural studies of the human enzyme. Kinetic constants (Km, kcat), exact substrate ranking, and the quaternary state have not been measured for the *P. putida* protein itself.
2. **Quaternary structure is inferred, not observed.** The homohexameric double-trimer assembly is documented for *E. coli*/*Salmonella* BCATs; while Q88H54 is an *E. coli*-type ortholog, its oligomeric state has not been experimentally determined.
3. **Residue numbering for the catalytic lysine is alignment-based.** Lys183 is assigned by pairwise alignment to *E. coli* IlvE; a crystal structure or PLP-labeling experiment on Q88H54 would confirm the identity of the Schiff-base lysine directly.
4. **In vivo directionality and flux control in *P. putida* are not directly measured.** The dual biosynthetic/catabolic role is inferred from pathway topology and from regulation of the

flanking *bkd* operon; the actual flux direction under given growth conditions in KT2440 has not been quantified.

5. **Redundancy in *P. putida* is unquantified.** The *L. lactis* knockout shows partial redundancy for Leu/Met/Phe; whether *P. putida* possesses additional aminotransferases that back up IlvE for these substrates is not established.
6. **Localization is predictive.** Cytoplasmic localization is inferred from the absence of a signal peptide/transmembrane region and from homolog studies, not from experimental fractionation of *P. putida*.

## Proposed Follow-up Experiments / Actions

1. **Recombinant expression and steady-state kinetics.** Express Q88H54 in *E. coli*, purify, and measure  $K_m/k_{cat}$  for each branched-chain 2-oxo acid and amino acid, plus 2-oxoglutarate/glutamate, to confirm the predicted substrate ranking and the minor Phe/Tyr activity directly for the *P. putida* enzyme.
2. **Cofactor and mechanism confirmation.** Verify PLP binding spectroscopically (characteristic ~415 nm internal aldimine), demonstrate the PLP↔PMP cycle, and confirm ping-pong kinetics by initial-velocity analysis.
3. **Site-directed mutagenesis of Lys183.** Mutate the predicted Schiff-base lysine (K183A) and show loss of activity, directly validating the alignment-based catalytic residue assignment.
4. **Quaternary structure determination.** Use size-exclusion chromatography with multi-angle light scattering (SEC-MALS) and/or crystallography/cryo-EM to test whether Q88H54 forms the *E. coli*-type homohexamer.
5. **Gene knockout in *P. putida* KT2440.** Construct a  $\Delta ilvE$  ( $\Delta PP\_3511$ ) strain and test (a) BCAA auxotrophy (biosynthetic role), (b) inability to grow on Val/Ile/Leu as sole nitrogen/carbon source (catabolic role, coupling to Bkd), and (c) residual transaminase activity toward each substrate to quantify redundancy.
6. **Subcellular localization.** Confirm cytoplasmic localization by cell fractionation and/or a functional fluorescent fusion in KT2440.
7. **Flux/regulation analysis.** Combine transcriptomics/proteomics with  $^{15}N/^{13}C$  metabolic-flux analysis under BCAA-biosynthetic versus BCAA-catabolic growth conditions to determine in vivo reaction directionality and how IlvE flux couples to *bkd* operon induction and Crc-mediated catabolite repression.

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*Report prepared for the functional annotation of ilvE (UniProt Q88H54, PP\_3511) in Pseudomonas putida KT2440. Target identity was verified against the UniProt reference; gene symbol, organism, protein family (class-IV PLP-dependent), and domain architecture all converge on a single, unambiguous BCAT annotation, with no gene-symbol ambiguity detected. All quoted statements are verbatim from the cited PubMed abstracts; sequence identity and residue mapping were computed from UniProt sequences Q88H54 and P0AB80.*

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